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Oefeningenreeks 3C nr. 12

Bewijs: als $0 < b \leq a$, dan geldt $\frac{a-b}{a} \leq \ln\frac{a}{b} \leq \frac{a-b}{b}$

- f is differentieerbaar in $]b, a[$
- f is continu in $[b, a]$

met $f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$

$$\overset{Lagrange}{\Rightarrow} f'(c) = \frac{f(a) - f(b)}{a-b} = \frac{\ln a - \ln b}{a-b} = \frac{\ln\frac{a}{b}}{a-b}$$

$$\Rightarrow f'(c) = \frac{\ln\frac{a}{b}}{a-b} = \frac{1}{c} \Leftrightarrow \frac{a-b}{c} = \ln\frac{a}{b} \text{ met } (b \leq c \leq a)$$

$$\Leftrightarrow \frac{1}{a} \leq \frac{1}{c} \leq \frac{1}{b}$$

$$\Leftrightarrow \frac{1}{a} \leq \frac{\ln\frac{a}{b}}{a-b} \leq \frac{1}{b}$$

$$\Leftrightarrow \frac{a-b}{a} \leq \ln\frac{a}{b} \leq \frac{a-b}{b}$$

References